

Sustainable Software Engineering - Course Introduction



Sustainable Software Engineering: General Information

Teachers

Gemma Catolino and Dario Di Nucci

Lectures Schedule

Monday: 13.30–15.30 Room F5

Tuesday: 13.30–15.30 Room F5

E-learning:

<https://elearning.informatica.unisa.it/el-platform/course/view.php?id=1270>



Gemma Catolino
gcatolino@unisa.it

Office Hours

Tuesday: 09:00 – 11:00

Thursday: 09:00 – 11:00



Dario Di Nucci
ddinucci@unisa.it

Office Hours

Wednesday: 14:00 – 16:00

Friday: 09:00 – 11:00

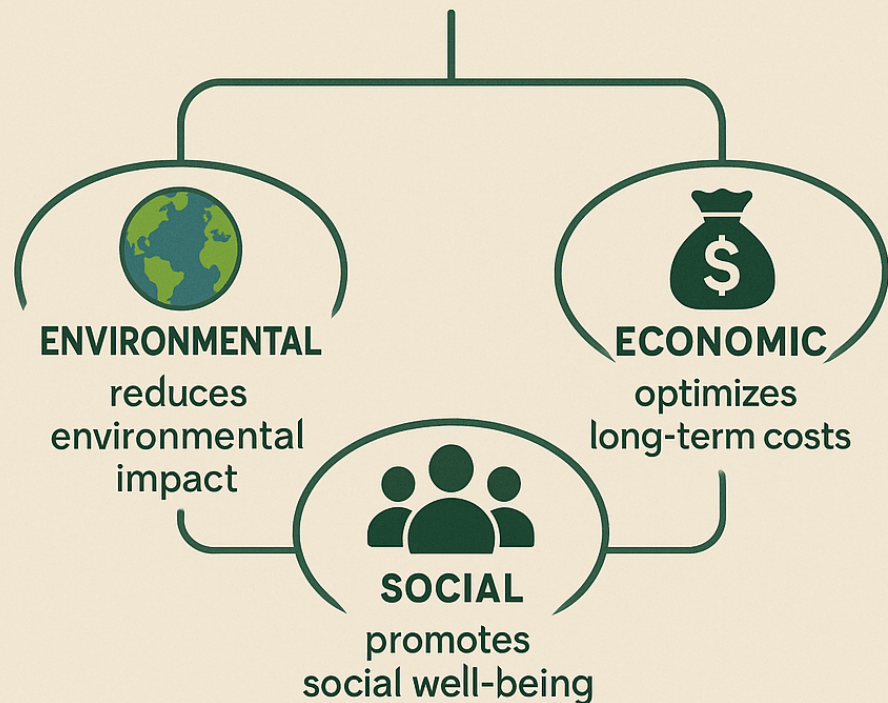


Please send an email at least one day before. The meeting can also be held using MS Teams.



Sustainable Software Engineering: Motivation behind this course

WHY SUSTAINABILITY IS IMPORTANT



Environmental impact of software

The ICT sector accounts for an estimated 2–3% of global CO₂ emissions – roughly the same as the airline industry.

Rising demand for sustainable software

Tech giants such as Google, Microsoft, and Meta have pledged to achieve carbon neutrality by 2030.

Economic advantages

Embracing sustainable development practices can lower operating costs while improving energy efficiency.

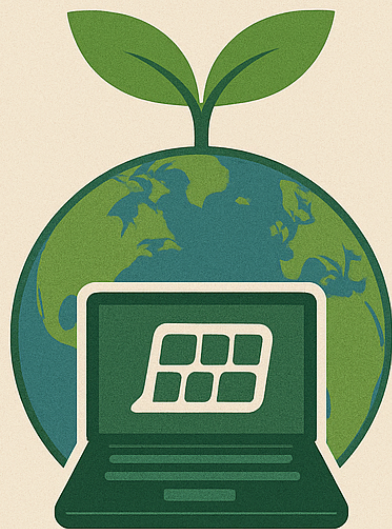
Developers' social responsibility

Every line of code carries an environmental footprint.

Sustainable Software Engineering: Goal of this course

Provide a solid understanding of sustainability in software engineering, focusing on energy efficiency, environmental, social, and economic impact.

SUSTAINABLE SOFTWARE ENGINEERING



Energy optimization

Teach how to develop software that minimizes energy consumption and reduces environmental impact.

Long-term costs

Design solutions that are both cost-effective and sustainable over time, lowering maintenance expenses.

Social impact

Train developers to create technologies that are inclusive and socially responsible.

Regulatory compliance

Prepare to address the growing regulations on sustainability and accountability in software.

Sustainable Software Engineering: Learning Outcomes

Knowledge & Understanding

Apply principles and tools to optimize software for sustainability.

Applying Knowledge

Design sustainable applications across the software lifecycle with a holistic approach.

Judgement

Make responsible design decisions considering technical, ethical, social, and environmental factors.

Communication

Clearly justify and present sustainable solutions to technical & non-technical audiences.

Learning Skills

Stay updated with emerging trends and apply sustainable practices in professional careers.

Sustainable Software Engineering: Course Overview

M1. Introduction to sustainability in software engineering

Presentation of basic concepts and the “triple” dimension of sustainability (environmental, social, economic) to provide the necessary theoretical foundations.

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M5. The role of developers and human factors in sustainability

Analysis of organizational culture, team well-being, and sustainability-oriented DevOps practices.

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Presentation of total cost of ownership (TCO) models and evaluation of architectural decisions from economic and equity perspectives.

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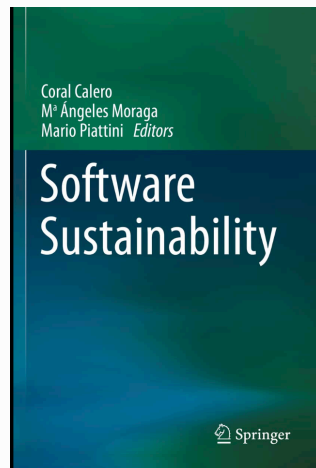
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M7. Case studies / Project

Review of real-world cases in green software and inclusivity.

Sustainable Software Engineering: Tentative Schedule

Module	Date	Topic
Introduction to sustainability in software engineering	23/09/2025	Course Introduction and Schedule + Sustainability 101
	29/09/2025	Sustainability (Three Sides) 101
Energy efficiency in software	30/09/2025	Energy efficiency and its impact
	06/10/2025	Measuring energy consumption
	07/10/2025	Energy patterns atelier & energy regression testing
Energy efficiency in software / Patterns and tools for sustainability	13/10/2025	Lab on energy monitoring and energy patterns: energybridge, e_regression, ecocode, greenit_analysis
	14/10/2025	Performance testing
	20/10/2025	Lab on unit-level performance testing: JHM, JU2JMH, AMBER
	21/10/2025	Lab on system-level performance testing: JMeter
Fairness and social justice in software	27/10/2025	Fairness 101
	28/10/2025	Fairness 101 - SoA
	03/11/2025	Lab on tool for measuring and mitigating Fairness
	04/11/2025	Lab on applying tools
The role of developers and human factors in software sustainability	10/11/2025	Diversity of the Teams and its impact
	11/11/2025	Anti pattern in Teams and social debt
	17/11/2025	Lab on community smells and socio-technical metrics
	18/11/2025	Lab on applying tools
Economic, ethical, and social evaluation of development choices	24/11/2025	Total cost ownership and technical debt
	25/11/2025	Software licensing
	01/12/2025	Lab on license compliance: FOSSA
Case studies / Project	02/12/2025	TBD
	09/12/2025	TBD
	15/12/2025	TBD
	16/12/2025	TBD

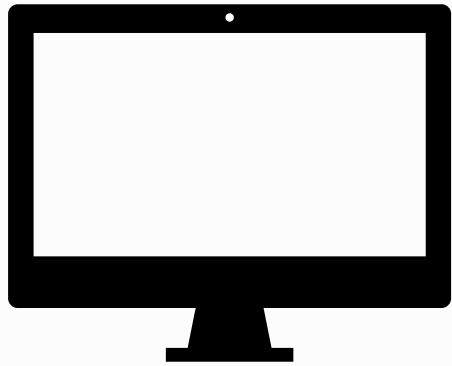


Calero, Coral, M^a Ángeles Moraga, and Mario Piattini.
"Introduction to software sustainability." *Software Sustainability*.
Cham: Springer International Publishing, 2021. 1-15.



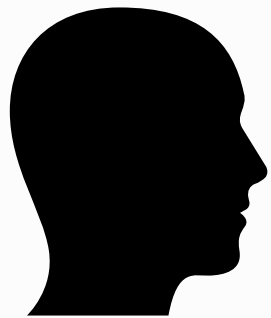
Slides provided by the teachers

Sustainable Software Engineering: Final Evaluation



Project

It will be the core of the exam. The grade will account for 50% of the final mark. The project must be presented and passed before the oral exam.



Oral Exam

It will cover all aspects of the course and will consist of 3-4 questions.

**Complete the
Project**

**Project
Discussion**

**Oral
Exam**



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